

Epic 6: Fitting the Model

Fit the Linear Regression Model

Linear Regression is a supervised machine learning algorithm.

It predicts continuous numerical values.

The model learns from the training dataset.

Independent variables are used as inputs.

The dependent variable is the prediction target.

The algorithm finds the best-fit line.

It minimizes prediction error.

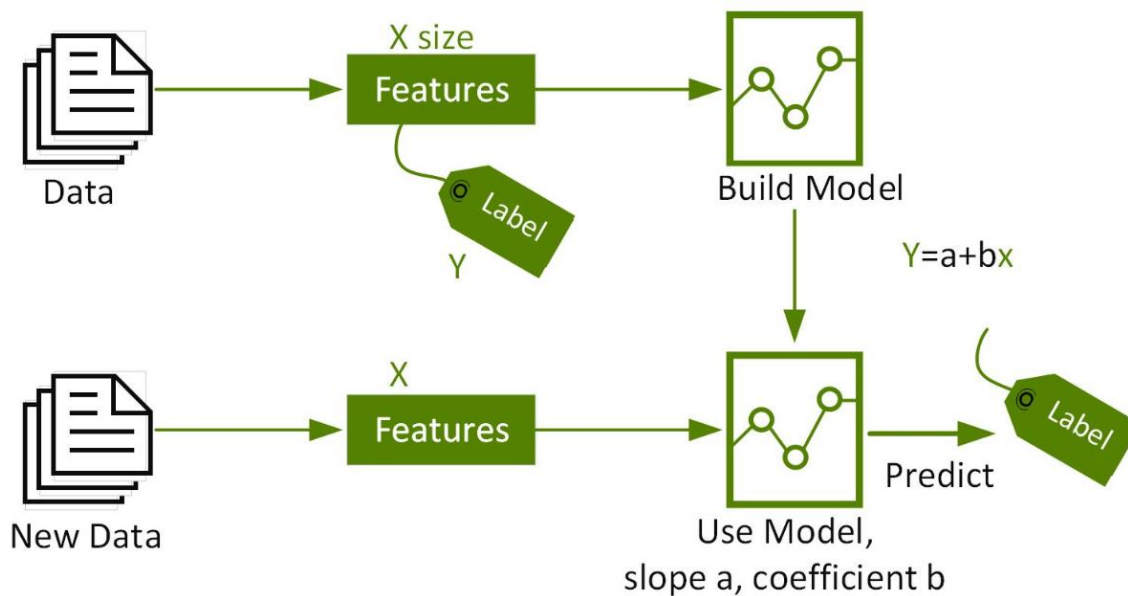
Model fitting identifies relationships among variables.

For the HDI project, indicators predict the Human Development Index.

The fitted model stores coefficients.

Training is completed before prediction.

A well-fitted model improves accuracy.



Working of Linear Regression

Training data is provided to the algorithm.

The model calculates coefficients.

It establishes a mathematical relationship.

The best-fit line minimizes squared errors.

Model parameters are optimized.

The trained model is saved.

Feature importance can be analyzed.

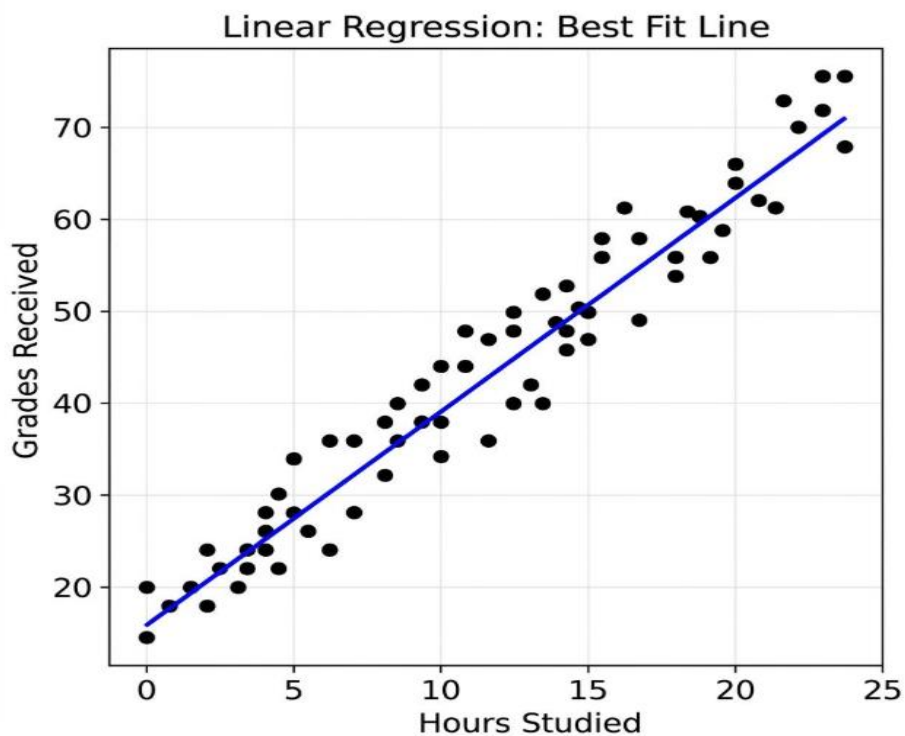
Proper preprocessing improves fitting.

The model should avoid overfitting.

Validation confirms performance.

Linear regression is simple and effective.

It is widely used in prediction tasks.



Predicting the Results

The trained model predicts values for test data.

Input features are passed to the model.

Predicted values are compared with actual values.

Performance metrics are calculated.

Prediction accuracy indicates model quality.

Errors help improve the model.

Results are visualized using graphs.

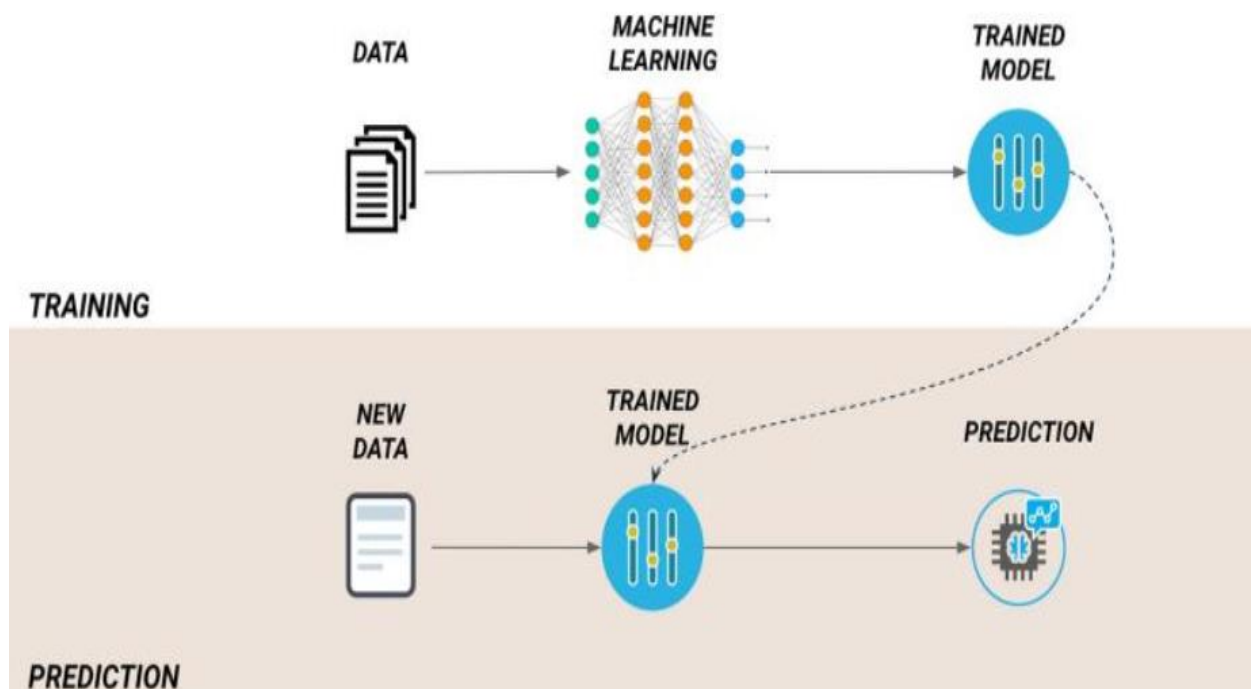
The HDI value is estimated for new data.

Reliable predictions support decision-making.

Testing validates the model.

The final output is generated.

This completes the machine learning workflow.



Linear Regression Process

Data Points



Plotted
observations

Best Fit Line



Line of best fit

Error Measurement



Calculate the residuals

Prediction



Make future
estimates